

CS 463: Cloud Native Platform Engineering

Course Syllabus

Instructor

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Student Learning Information

Course Description

Introduction to various aspects of software design, development and architecture used to create modern cloud computing software products, that run cost-effectively at scale. Focus will be placed on covering software engineering concepts, techniques and technologies used to build and deploy applications that run at scale on cloud infrastructure. Key topics include cloud native platform engineering concepts such as developing and deploying infrastructure to support a modern API-based application in the cloud.

College/Department: CCI, Department of Computer Science

Repeat Status: Not repeatable for credit

Prerequisites: CS 283 – Systems Programming [Min Grade: C]. The ideal student will have background in application design and development along with systems programming concepts including some familiarity with operating systems.

Credits: 3 hours of lecture (3 credits total)

Course Purpose within a Program of Study

This undergraduate-level course is an elective for BS CS and BS SE degree programs

Statement of Expected Learning

The course objectives are to:

- Study how cloud infrastructure enables software products to be deployed with resiliency and run at scale
- Concepts associated with treating the Cloud as nothing more than a “special purpose operating system”
- Explore patterns and implications associated with designing, deploying and coordinating ephemeral software components (aka serverless)
- Explore cloud runtime application platforms including virtual machines, container engines and serverless solutions.
- Exploring APIs including how to build them, and software engineering patterns such as event-driven architectures that are commonly used in the cloud
- Investigating common tooling, programming languages and frameworks used to create modern cloud native systems
- Investigating the 6-Pillars of well-architected cloud systems – Operational excellence, Security, Reliability, Performance efficiency, Cost optimization, and Sustainability.

As learning outcomes, students completing this course should be able to:

- Understand modern concepts associated with working with the cloud and building simple applications using modern cloud infrastructure. These include, VPCs, Cloud Networking, Cloud Security
- Software architectures, frameworks and patterns used to construct resilient and scalable microservices and modern API based software products
- Have a solid understanding of cloud platform architecture concepts such as regions, availability zones, edge locations
- Obtain hands on experience developing and deploying a simple API to cloud infrastructure.

What this course is not about:

This course is not intended to go deep into a particular cloud providers offering for the purposes of achieving cloud certification. There are many excellent resources elsewhere that focus on the specifics of these cloud provider platforms for the purposes of achieving certification. While this course does not focus on “how to do things” deeply using a specific cloud providers suite of services, this course instead is designed to address an orthogonal concern associated with how to design and deployed well-engineered cloud systems.

Course Time Commitment Disclosure

The time commitment required to successfully complete this course will vary significantly depending on students experience with software development toolchains, programming languages and build tools. If you have any questions or doubts about your readiness to take this course, please setup time with the instructor to discuss.

Course Materials

Required and Recommended Texts, Readings, and Resources

There is no required textbook for this course. Materials will be provided by the instructor in Blackboard, and complemented with free resources to support labs and assignments by a leading public cloud provider (TBD: AWS or GCP). The instructor will also provide a GitHub repository containing demonstration- and starter- code for various course deliverables.

Assignments, Assessments, and Evaluations

Graded Assignments and Learning Activities

This course will balance hands on application with more traditional homework assignments. A course project will be assigned with multiple deliverables that follow the course lectures. In addition, there will be approximately 5-6 written assignments involving reviewing material provided by the instructor (research papers, videos, etc) and answering questions based on specific prompts.

Due Dates and your Late Day Bank

This course will have assignments that will be spread out over the course as mentioned above. Given my previous experience, students tend to have different levels of readiness, especially with background in programming. I will adjust deadlines for the whole class as

necessary, based on my observation of student effort and questions being asked over Discord, and attending office hours (mine or the TAs). One off deadline extensions are not possible, but I do realize that things come up from time to time. Prior to the class starting I will populate a column in Blackboard grade center called your "late day bank" with 5 late days to use over the term. Late submissions will be accepted without penalty by deducting from your late day bank. Once you exhaust your bank, late assignments will receive a 50% penalty on day 1 and will not be accepted on day 2+. Assignments are typically due at the end of the week (midnight on Friday).

Students are responsible for checking Blackboard Learn and Drexel email daily for course announcements and changes.

For more details, please refer to The Drexel University Student Handbook.

Expectations of Timely Communication for Hardships

If, for whatever reason, you find yourself struggling with keeping up with course deliverables, it is important that you let me know right away, including your plan to get back on track. I am here to help you and am willing to be as flexible as possible. However, my flexibility becomes very limited if you come to me weeks after one or more assignment deadlines have been missed with a hardship story expecting me to accept missing work without significant penalty, if even at all.

Grading Matrix

Grades will be assigned based on the following:

- Homework Assignments and Hands-On Lab Deliverables (see weekly schedule): 60%.
- Bi-Weekly online quizzes: 40%

More on quizzes and hands-on deliverables:

- The first half of this course will consist of a series of independent deliverables as students learn about cloud concepts, the second half of the course will have deliverables that build up to a final course project deliverable.
- This course will include 5 online quizzes (approximately every-other-week). Your quiz grade will be determined as follows:

5 quizzes @ 25 points each

Final Quiz Grade = (4 highest quiz scores * 25) + (0.5 lowest quiz score)
Essentially your lowest quiz score counts as extra credit.

Grade Scale

The following scale will be used to convert points to letter grades:

Points	Grade	Points	Grade	Points	Grade
Exceptional	A+	83-86.99	B	70-72.99	C-
93-99.99	A	80-82.99	B-	67-69.99	D+
90-92.99	A-	77-79.99	C+	60-66.99	D
87-89.99	B+	73-76.99	C	0-59.99	F

Note that the instructor may revise this conversion if/when necessary.

Course Schedule

[This schedule is tentative and may change during the course.]

This course is designed to balance both theory and practical application. Weekly activities will be split between traditional lectures and hands on in-class activity labs. Student will be expected to create and maintain a Drexel GitHub repository to

1. Course Introduction & Introduction to Cloud Computing
In Class Activity and Labs: None, Homework: Summarize instructor provided video.
2. Cloud Computing Models and Platforms – IaaS, PaaS, FaaS
Fully managed Cloud Services
In Class Activity / Lab: Creating and deploying a simple web server in the cloud
3. Cloud Platform Engineering – Regions, Availability Zones, API-Driven Platforms
In Class Activity / Lab: Creating and Securing a simple VPC (virtual private cloud)
Quiz 1 Due
4. Cloud Networking – Creating and securing custom network topologies within VPCs that support high availability.
In Class Activity / Lab: Expand VPC from week 3. Create a VPC-based network that includes both private and public subnets. Configuration of Network ACLs and Routing between the public and private zones. Configuration of a Bastion Host (jump box) to ssh into private infrastructure.
5. API-Based Architectures. What is an API, how do we build them, and some common API based patterns (request/response, event-driven).
In Class Activity / Lab: Build and test a simple API on your local laptop, then redeploy on virtual machine in the cloud.
Quiz 2 Due
6. Cloud Native Application Architectures – focus on the AWS well architected framework.
In Class Activity / Lab: *Cloud Native Design Challenge (Simple API-based Application) using concepts from the well architected framework.*
7. Introduction to Serverless – aka Function as a Service (FaaS) and API Gateways
In Class Activity / Lab: Final Project Part 1: Refactor and redeploy simple API from week 5, as a serverless API.
Quiz 3 Due
8. Cloud Native Storage and Databases – introduction to cloud native database concepts examining both traditional SQL-based, and cloud native document and KV solutions offered by cloud providers.
In Class Activity / Lab: Final Project Part 2: Expand API from week 7 and add a persistence layer to read and write data to a cloud native database.
9. Part 1: Advanced Cloud Native Computing Concepts – Containerization and Container based Runtimes
In Class Activity / Lab: Final Project Part 3: Add external API integration to the API you have been working on.
Quiz 4 Due

10. Part 2: Advanced Cloud Native Computing Concepts – High Availability: Auto Scaling Groups, Multi AZ topologies, etc.

In Class Activity / Lab: Finalize course project, create project documentation and a test suite using postman and or insomnia.

11. No class, no final exam – Final Project Due (link to course project git repo + short personal video demonstrating what you created)

Quiz 5 Due

Diversity Statement

CCI's vision is to nurture an inclusive community for learning and working where differences are valued, and each person is supported and empowered to reach their full potential, making CCI a powerful enabler of an equitable and just society. For more information, please see <https://drexel.edu/cci/about/diversity-equity-and-inclusion-council/>

Academic Policies

This course follows university, college, and department policies, including but not limited to:

- Academic Integrity, Plagiarism, Dishonesty and Cheating Policy:
<https://drexel.edu/provost/policies-calendars/policies/academic-integrity/>
- Students with Disability Statement:
<https://drexel.edu/disability-resources/support-accommodations/student-family-resources/>
- Course Add-Drop Policy:
<https://drexel.edu/provost/policies-calendars/policies/course-add-drop/>
- Course Withdrawal Policy:
<https://drexel.edu/provost/policies-calendars/policies/course-withdrawal/>
- CCI Academic Affairs Policies:
<https://drexel.edu/cci/current-students/policies/>
- Drexel Student Learning Priorities:
<http://drexel.edu/provost/assessment/outcomes/dslp/>

The instructor(s) may, at his/her/their discretion, change any part of the course before or during the term, including assignments, grade breakdowns, due dates, and schedule. Such changes will be communicated to students via the course web site. This web site should be checked regularly and frequently for such changes and announcements.

Students [requesting accommodations](#) due to a disability at Drexel University need to request a current Accommodations Verification Letter (AVL) in the [ClockWork database](#) before accommodations can be made. These requests are received by Disability Resources (DR), who then issues the AVL to the appropriate contacts. For additional information, visit the DR website at <http://drexel.edu/oed/disabilityResources/overview/> , or contact DR for more information by phone at 215.895.1401, or by email at disability@drexel.edu.